$\det \mathscr{G}_h' = 1$. Thus the asymmetric variant of the Euler method is area preserving, even though the standard Euler method is not.

It would be valuable to have a general method for deriving volume conserving methods. It turns out that volume conservation is, itself, most readily obtained as a consequence of a more fundamental property of Hamiltonian flows, the conservation of the symplectic form.

2.3.3 The Symplectic Property

Let $m = 2\mathrm{N}_c$. A symplectic map $\boldsymbol{\Phi} : \mathbb{R}^m \to \mathbb{R}^m$ is one that preserves the symplectic differential 2-form. The simplest way to write this is as the following algebraic condition on the Jacobian matrix of $\boldsymbol{\Phi}$:

$$\boldsymbol{\Phi}'^T \boldsymbol{J} \boldsymbol{\Phi}' = \boldsymbol{J}.$$

Recall that the Jacobian matrix is the $m \times m$ matrix of partial derivatives of the components of $\boldsymbol{\Phi}$, $(\boldsymbol{\Phi}')_{ij} = \frac{\partial \Phi_i}{\partial z_j}$, while

$$\boldsymbol{J} = \begin{bmatrix} \mathbf{0} & \boldsymbol{I} \\ -\boldsymbol{I} & \mathbf{0} \end{bmatrix}.$$

So the equation $\boldsymbol{\Phi}'^T \boldsymbol{J} \boldsymbol{\Phi}' = \boldsymbol{J}$ represents a large number ($4\mathrm{N}_c^2$) of equations.

To explain the origin of this formula, we need a brief diversion into the world of differential forms. This language will also be useful in Chap. 4, where we consider constrained systems.

A *1-form* α defined on $\mathbb{R}^m$ is a family of linear mappings from $\mathbb{R}^m$ to $\mathbb{R}$, defined for each point of $\mathbb{R}^m$. Let $\boldsymbol{a} : \mathbb{R}^m \to \mathbb{R}^m$, then we may define a one-form associated to this vector by $\alpha(\boldsymbol{x})(\boldsymbol{\xi}) = \boldsymbol{a}(\boldsymbol{x})^T \boldsymbol{\xi}$.

The *differential* of a function $g : \mathbb{R}^m \to \mathbb{R}$, denoted $\mathrm{d}g$, is a family of linear mappings (one for each point in phase space) from vectors $\boldsymbol{\xi} \in \mathbb{R}^m$ into the reals defined by

$$\mathrm{d}g(\boldsymbol{q}, \boldsymbol{p})(\boldsymbol{\xi}) = \nabla g(\boldsymbol{q}, \boldsymbol{p})^T \boldsymbol{\xi}.$$

So, denoting the ith position coordinate by q_i, we have $\mathrm{d}q_i(\boldsymbol{\xi}) = \xi_i$; the differential is thus an example of a 1-form.

On the other hand the so called *wedge product* of 1-forms α, β, is an example of a *2-form*, at any point in phase space it can be viewed as a quadratic form, i.e. a scalar valued function of two vectors which is linear in each argument. It is written $\alpha \wedge \beta$ and is defined, for vectors $\boldsymbol{\xi}, \boldsymbol{\eta} \in \mathbb{R}^m$ by

$$\alpha \wedge \beta(\boldsymbol{\xi}, \boldsymbol{\eta}) = \alpha(\boldsymbol{\xi})\beta(\boldsymbol{\eta}) - \alpha(\boldsymbol{\eta})\beta(\boldsymbol{\xi}).$$